

Online Shopping Management System using Java OOP

1. Project Overview

The Online Shopping Management System is a console-based Java application designed to simulate a real e-commerce platform. The system allows customers to browse products, add items to cart, purchase products, manage balances, and view order history.

The project demonstrates the four major Object-Oriented Programming (OOP) concepts in Java:

1. Encapsulation
2. Inheritance
3. Polymorphism
4. Abstraction

The application uses:

- Classes and Objects
- Constructors
- ArrayList
- Loops
- Conditions
- Switch statements
- Method Overriding
- Abstract Classes

The project is designed to behave like a small real shopping application with a menu-driven system.

2. Objectives of the Project

The main objectives of this project are:

- Apply OOP concepts in a real-world application
 - Build a menu-driven console application
 - Practice Java collections and methods
 - Simulate shopping system operations
 - Improve problem-solving and coding structure
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3. OOP Concepts Used

3.1 Encapsulation

Encapsulation is implemented by making class attributes private and accessing them using getters and setters.

Example:

```
private double price;
```

```
public double getPrice() {  
    return price;  
}
```

Used in:

- Product class
- Customer class
- Order class

3.2 Inheritance

Inheritance is used to create different user types.

Example:

```
class PremiumCustomer extends Customer
```

```
class RegularCustomer extends Customer
```

Both classes inherit common customer properties.

3.3 Polymorphism

Polymorphism is implemented using method overriding.

Example:

```
@Override
```

```
public void calculateDiscount()
```

Each customer type has different discount logic.

3.4 Abstraction

Abstraction is implemented using an abstract class.

Example:

```
abstract class Customer {  
    abstract void calculateDiscount();  
}
```

4. System Features

The application provides the following features:

1. Add Product
 2. Display Products
 3. Search Product
 4. Add Product to Cart
 5. Remove Product from Cart
 6. View Shopping Cart
 7. Purchase Products
 8. Add Balance
 9. View Balance
 10. Generate Invoice
 11. Display Order History
 12. Customer Information
 13. Apply Discounts
 14. Exit System
-

5. Classes Used in the Project

5.1 Product Class

Responsibilities:

- Store product information
- Manage stock quantity

Attributes:

- productId
- productName
- price
- quantity

Methods:

- displayProduct()
 - updateQuantity()
-

5.2 Abstract Customer Class

Responsibilities:

- Store customer data
- Manage balance

Attributes:

- customerId
- customerName
- balance

Methods:

- addBalance()
 - displayCustomerInfo()
 - abstract calculateDiscount()
-

5.3 RegularCustomer Class

Inheritance:

- extends Customer

Features:

- Standard shopping operations
 - Small discount percentage
-

5.4 PremiumCustomer Class

Inheritance:

- extends Customer

Features:

- Higher discounts
 - Special offers
-

5.5 Cart Class

Responsibilities:

- Add products
- Remove products
- Calculate total price

Methods:

- addToCart()
 - removeFromCart()
 - viewCart()
 - calculateTotal()
-

5.6 Order Class

Responsibilities:

- Generate invoices
- Store purchased items

Methods:

- createOrder()
 - displayInvoice()
-

5.7 ShoppingSystem Class

Responsibilities:

- Manage the whole application
 - Control products and users
-

6. System Workflow

Step 1

The application starts.

Step 2

The main menu appears.

Step 3

The user selects an operation.

Step 4

The system processes the selected action.

Step 5

The menu keeps appearing until the user exits.

7. Main Menu Example

===== Online Shopping System =====

1. Add Product
2. Display Products
3. Search Product
4. Add Product To Cart
5. Remove Product From Cart
6. View Cart
7. Purchase Products
8. Add Balance
9. View Balance
10. Generate Invoice
11. View Order History
12. Customer Information
13. Apply Discount
14. Exit

Enter Your Choice:

8. Java Classes

Product Class

```
class Product {  
  
}
```

Abstract Customer Class

```
abstract class Customer {  
  
}
```

PremiumCustomer Class

```
class PremiumCustomer extends Customer {  
  
}
```

RegularCustomer Class

```
class RegularCustomer extends Customer {  
  
}
```

Cart Class

```
import java.util.ArrayList;  
  
class Cart {  
  
    ArrayList<Product> cartItems = new ArrayList<>();  
  
    public void addToCart(Product product) {  
  
    }  
  
    public void removeFromCart(String name) {  
  
    }  
}
```

```
    public void viewCart() {  
  
    }  
  
    public double calculateTotal() {  
  
    }  
}
```

Main Class

```
import java.util.ArrayList;  
import java.util.Scanner;  
  
public class Main {  
  
    public static void main(String[] args) {  
    }  
}
```

9. Expected Output Example

===== Shopping System =====

1. Add Product
2. Display Products
3. Add Product To Cart
4. View Cart
5. Purchase
6. Add Balance
7. View Customer Info
8. Exit

Enter Choice: 1

Enter Product ID: 101

Enter Product Name: Laptop

Enter Price: 1200

Enter Quantity: 5

Product Added
